

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June R 2024 Worked Solutions

May/June R 2024 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P2

PAST PAPER

WMA12/01 May/June R 2024

May/June R 2024 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Sequences & Series

1. The sequence $u_1, u_2, u_3, \dots$ satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

- (a) find the value of u_2

(2)

- (b) find $\sum_{r=1}^4 (u_r + 2r)$

(3)

(Total for Question 1 is 5 marks)

(Total for Question 1 is 5 marks)

Worked Solution - Question 1

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: $u_1 = a$, $u_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d)$ or $S_n = \frac{n}{2}(u_1 + u_n)$.

Final answer

$u_1 = a$, $u_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d)$ or $S_n = \frac{n}{2}(u_1 + u_n)$.

Question 2

Integration

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

(Total for Question 2 is 5 marks)

(Total for Question 2 is 5 marks)

Worked Solution - Question 2

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $\times = 11$

Final answer

$\times = 11$

Question 3

Circles

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is nearest the origin O ,

(b) find the exact length of OP

(2)

(Total for Question 3 is 5 marks)

(Total for Question 3 is 5 marks)

Worked Solution - Question 3

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: 12 (b)

Final answer

12 (b)

Question 4

Binomial Expansion

4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

- (b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

(Total for Question 4 is 7 marks)

(Total for Question 4 is 7 marks)

Worked Solution - Question 4

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: 738 but

Final answer

738 but

Question 5

Trigonometric Equations

5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

- (a) Show that the depth of water in the harbour at 2 am is just over 4 metres. (1)
- (b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres. (5)

(Total for Question 5 is 6 marks)

(Total for Question 5 is 6 marks)

Worked Solution - Question 5

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: d 13:04 or 1:04 pm

Final answer

d 13:04 or 1:04 pm

Question 6

Applications of Differentiation

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where p and q are constants.

Given that

- $(2x + 3)$ is a factor of $f(x)$
- $f(x)$ has a remainder of -5 when divided by $(x + 2)$

(a) (i) show that $p = 10$

(ii) find the value of q .

(5)

(b) Hence find the range of values of x for which $f(x)$ is decreasing.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 6 is 9 marks)

(Total for Question 6 is 9 marks)

Worked Solution - Question 6

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 7

Trigonometric Equations

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

- (ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

(Total for Question 7 is 7 marks)

(Total for Question 7 is 7 marks)

Worked Solution - Question 7

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: θ

Final answer

θ

Question 8

Arithmetic Sequences

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} \{2a + (n-1)d\} \quad (3)$$

- (b) Hence find

$$900 + 892 + 884 + \dots + 500 \quad (3)$$

- (ii) Given that the first three terms of a **geometric** series are

$$k+4 \quad k-2 \quad 11-k$$

where k is a constant,

- (a) show that

$$2k^2 - 11k - 40 = 0 \quad (3)$$

Given also that this series is convergent,

- (b) find the value of S_∞ (4)

(Total for Question 8 is 13 marks)

(Total for Question 8 is 13 marks)

Worked Solution - Question 8

1. Use the arithmetic formula

Identify the first term, common difference, and required term or sum from the question.

2. Substitute and solve

Use the arithmetic term or sum formula, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $- = + = +$

Final answer

$- = + = +$

Question 9

Laws of Logarithms

9. Given that

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,

(2)

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

giving each answer in simplest form.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

(Total for Question 9 is 9 marks)

(Total for Question 9 is 9 marks)

Worked Solution - Question 9

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: +

Final answer

+

Question 10

Applications of Differentiation

10.

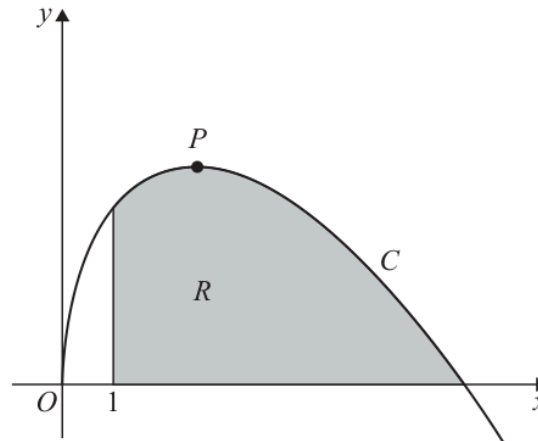


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point P is the stationary point on C .

- (a) Find, using calculus, the x coordinate of P .

(4)

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 1$

- (b) Using calculus, calculate the exact area of R .

(5)

(Total for Question 10 is 9 marks)

(Total for Question 10 is 9 marks)

Worked Solution - Question 10

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: 148 5

Final answer

148 5